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ORIGINAL ARTICLE

BSCAL: Gesture Recognition Network Based on Dual-Stream Information Fusion of EMG and IMU Signals

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ABSTRACT

Background: Surface electromyography (sEMG) enables gesture recognition for rehabilitation and human-computer interaction but faces challenges from noise, inter-subject variability, and limited motion dynamics characterisation.

Methods: We propose BSCAL, a dual-stream network integrating sEMG and inertial measurement unit (IMU) signals. The architecture combines convolutional neural networks (CNNs), long short-term memory (LSTM) networks, and spatiotemporal attention to process eight-channel sEMG and six-channel IMU data from 20 participants performing six gestures. Kalman filtering, Z-score normalisation, and gated fusion optimise multimodal feature integration.

Results: BSCAL achieved 90.41% ± 1.54% recognition accuracy, surpassing single-modality models and state-of-the-art approaches. Ablation studies validated contributions from CNN (local features), LSTM (temporal dependencies), and attention (key feature enhancement).

Conclusions: By synergistically integrating neuromuscular patterns from EMG and kinematic data from IMU, BSCAL delivers a precise and scalable solution for gesture recognition, thereby advancing the development of wearable hand rehabilitation robots and assistive technologies for motor function recovery.

1 | Introduction

Stroke is a leading cause of permanent disability [1]. Approximately 70% of survivors experience upper limb motor impairment, with severe deficits in hand motor function [2]. As the population ages, the incidence of stroke-induced hand dysfunction is expected to rise further, underscoring the urgent need for novel and effective rehabilitation technologies to enhance the recovery of affected limbs. Conventional rehabilitation therapies rely on manual assistance from physical therapists, which faces challenges such as low efficiency, difficulty in quantification, and lack of personalisation [3]. In recent years, advancements in medical robotics have introduced breakthroughs in rehabilitation medicine, among which bio-signal-based human-machine interaction systems have emerged as a key technology for improving rehabilitation outcomes [4].

Surface electromyography (sEMG) signals, captured via non-invasive electrodes, sensitively reflect subtle hand movement intentions and have been widely applied in rehabilitation medicine [5], movement intention detection [6], exoskeleton robots [7], rehabilitation robots [8], and intelligent prosthetics [9]. EMG signals play a critical role in decoding hand movement issues [10–12]. Studies on the relationship between finger movements and EMG have demonstrated the effectiveness of EMG signals in recognising hand movement information [13–16]. Reference [17] proposed a fusion model integrating Squeeze-and-Excitation Networks (SE) and DenseNet, incorporating an attention mechanism. This method was tested on the sEMG gesture datasets NinaPro DB2 and DB4, achieving accuracies of 85.93% and 82.39%, respectively. Moslhi et al. combined the Singer Transformer with an attention mechanism [18], improving signal classification accuracy. Lee et al. designed a classification model based on Graph Convolutional Network (GCN) [19], which effectively extracts and aggregates key features associated with various gestures. Islam et al. proposed a lightweight All-ConvNet + transfer learning (TL) model [20] to enhance gesture recognition performance across dialogues and topics. Chen Qingzheng et al. converted EMG signals into grayscale images and utilised a CNN model for classification tasks [21]. This approach demonstrated strong adaptability and high recognition accuracy for low sampling rates and multichannel EMG data. However, the process of converting EMG signals from the time domain to the spatial domain may result in the loss of important feature information. Moreover, EMG signals are susceptible to muscle fatigue and electrode displacement [22], limiting the reliability of their standalone applications. Meanwhile, inertial measurement units (IMUs) can capture kinematic information of limbs, but they still suffer from signal drift issues in complex gesture recognition [23, 24], leading to reduced accuracy. Therefore, a single sensor modality struggles to meet the dual requirements of control robustness and accuracy in rehabilitation robots.

Multimodal information fusion offers a viable approach to address the aforementioned challenges. Tryon et al. employed a weighted fusion method to integrate electromyography (EMG), electroencephalography (EEG), and velocity signals. They classified elbow flexion-extension movements under three different load conditions, achieving an accuracy of 83.01 $\pm$ 6.04% [25]. Reference [26] fused EMG and electrocardiography (ECG) signals at the feature level and designed an improved particle swarm optimization-support vector machine (PSO-SVM) classifier, which enhanced the recognition accuracy of EMG signals under fatigue conditions by 13.5% compared to non-fused signals. Researchers have integrated physical signals with bioelectrical signals to achieve more accurate recognition. Tao et al. [27] fused inertial measurement unit (IMU) signals from the upper limb with EMG signals and utilised a convolutional neural network (CNN) to recognise six common arm movements during industrial tasks. The fused feature information improved the recognition rates by 0.1% and 2.4% compared to using IMU or EMG signals alone, respectively. Reference [28] proposed a lightweight gesture recognition network integrating depthwise separable convolutions and gated recurrent units, achieving recognition accuracies of 92.03 $\pm$ 3.28% and 77.48 $\pm$ 4.38% on the NinaPro DB2 and DB5 datasets, respectively. Wei et al. extracted features from EMG signals [29], and utilised a multi-stream CNN to model multi-view EMG signals, dynamically adjusting feature selection strategies based on model performance to classify gestures, demonstrating the superiority of the multi-view framework. These studies indicate that fusing bioelectrical and physical motion signals can significantly enhance the discriminative capability of models in complex scenarios. Nevertheless, existing methods often convert sEMG into images or rely on handcrafted features, which may lead to information loss or high computational complexity, hindering embedded deployment and real-time rehabilitation applications.

To address the aforementioned challenges, this study proposes a multimodal fusion recognition framework for hand function training in rehabilitation robotics, with a focus on its clinical application potential in active-assistive control. During rehabilitation training, the system can accurately identify users' active movement intention. Starting from raw signals, this research aims to enhance the robustness and accuracy of gesture recognition in complex environments through deep fusion of sEMG and IMU data, thereby further improving the interactive intelligence and personalized assistance capabilities of rehabilitation robots. Ultimately, it serves the core medical objective of restoring or compensating for patients' hand functions.

This study designed a recognition system utilising raw EMG and IMU signals for active training modalities in hand rehabilitation robots, proposing a novel strategy for hand motion recognition through the integration of electrophysiological and kinematic signals. To recognise hand movements, we developed a multimodal hand motion recognition model incorporating Kalman filtering algorithms and a hierarchical independent CNN-LSTM neural network architecture. The system processes 8-channel EMG signals and 6-channel IMU data while integrating a spatiotemporal attention mechanism for finger movement recognition. This design enables effective processing of multimodal time-series data while enhancing feature extraction capabilities. The key innovations include the following:

Dual-Stream Signal Processing Architecture: A dual-stream feature encoding framework tailored for EMG and IMU signals. For EMG's non-stationary characteristics, we designed a hybrid structure combining deep convolutional networks with bidirectional LSTM. For the low-frequency motion features of IMUs, a lightweight architecture integrates single-layer convolution with shallow LSTM. A custom spatiotemporal attention module dynamically enhances critical sensor channels and temporal segments through channel-time dual-dimensional weight calibration during feature encoding.

Dynamic Gated Fusion Strategy: A learnable gated weighting mechanism for multimodal fusion. Features are first projected into a unified dimensional space via fully connected layers, followed by modality importance coefficient generation using sigmoid activation, enabling dynamic feature-level weighted fusion.

Multimodal Data Integration: Systematic fusion of EMG and IMU data (including accelerometer and gyroscope signals) achieves superior gesture recognition accuracy, experimentally validating the effectiveness of multimodal fusion in enhancing system performance.

The remainder of this paper is organised as follows. Section 2 presents the movement sources and data preprocessing techniques, as well as the architecture and parameters of the proposed BSCAL model. Section 3 comprehensively discusses the experimental results. Finally, Section 4 concludes with research contributions and outlines future directions.

2 | Materials and Methods

2.1 | EMG and IMU Signal Data Acquisition

Statement: Ethical approval for this study was obtained from the Ethics Committee of Northeast Forestry University. All participants gave written informed consent prior to the experiment.

This study recruited 20 healthy participants (10 males, 10 females; age: 24.6 $\pm$ 2.3 years), with evaluations and instrumentation administered by trained researchers. Each participant wore an 8-channel electromyography (EMG) armband (23 cm length, 123.7 g weight) that simultaneously acquired eight channels of EMG signals and six channels of inertial measurement unit (IMU) data. The EMG armband was positioned on the dorsal forearm 5 cm distal to the elbow joint (see Figure 1). Prior to data collection, participants maintained a relaxed stationary posture with the wrist, forearm, and elbow resting on a table to ensure accurate forearm angle estimation. The wearable device synchronously captured IMU and EMG data at sampling frequencies of 50 and 500 Hz, respectively. Participants performed computer-guided tasks on an experimental platform while wearing the armband. The movements were selected based on the hand function assessment items of the Fugl-Meyer Assessment (FMA) scale [30], including mass flexion/extension, cylindrical grasp, pincer grasp, hook grasp, spherical grasp, and lateral pinch (see Figure 2). These movements are widely adopted in clinical practice to evaluate basic motor control capabilities of stroke patients' hands (e.g., finger mobility, basic grasping, and fine pinching) as well as their functional performance in simulating Activities of Daily Living (ADLs). The selection aims to ensure clinical relevance and standardisation of the study, focusing on the most fundamental and representative hand function metrics in rehabilitation assessment.

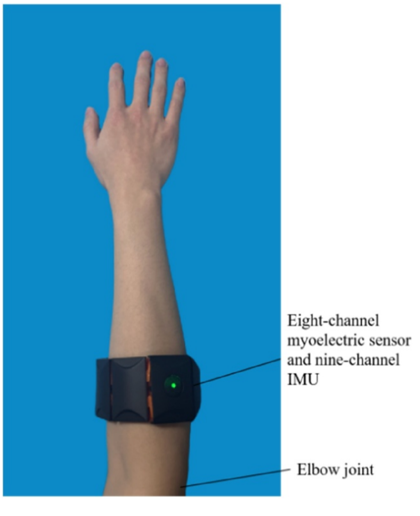

FIGURE 1 | EMG armband position.

2.2 | EMG and IMU Signal Processing

2.2.1 | Python-Based Software Implementation

PyCharm, a Python-integrated development environment (IDE) for scientific computing, was employed to implement the machine learning algorithms and deep learning models proposed in this study. The platform integrates libraries including NumPy and Matplotlib for dataset analysis and preprocessing, alongside TensorFlow 2.6 and Keras 2.6 for constructing classification and prediction architectures.

2.2.2 | Hand Motion Data Acquisition

The system integrates an inertial measurement unit (IMU) and surface electromyography (sEMG) sensors with temporal synchronization for precise hand motion feature acquisition. EMG sensors detect microvolt-level bioelectrical signals during muscle contraction, providing neuromuscular-level intent prediction 30–150 ms prior to movement execution. Concurrently, the IMU synchronously captures hand kinematic parameters (including acceleration a and angular velocity ω) via a three-axis accelerometer and gyroscope. The experiment employed a wearable data acquisition system, where multimodal signals were transmitted to a host computer through a serial port. The dataset comprises 1200 samples of 8-channel EMG signals (20 subjects $\times$ 6 gestures $\times$ 10 trials) and 1200 samples of 6-channel IMU signals (20 subjects $\times$ 6 gestures $\times$ 10 trials). To enhance system robustness, 10-fold cross-validation was adopted for model training. This method efficiently utilises limited data by iteratively training and validating the model, reducing performance fluctuations caused by biased single data partitioning and thus providing a more reliable evaluation of generalisation capability. Additionally, it facilitates optimization of hyperparameter selection, ensuring the stability and reliability of the final model.

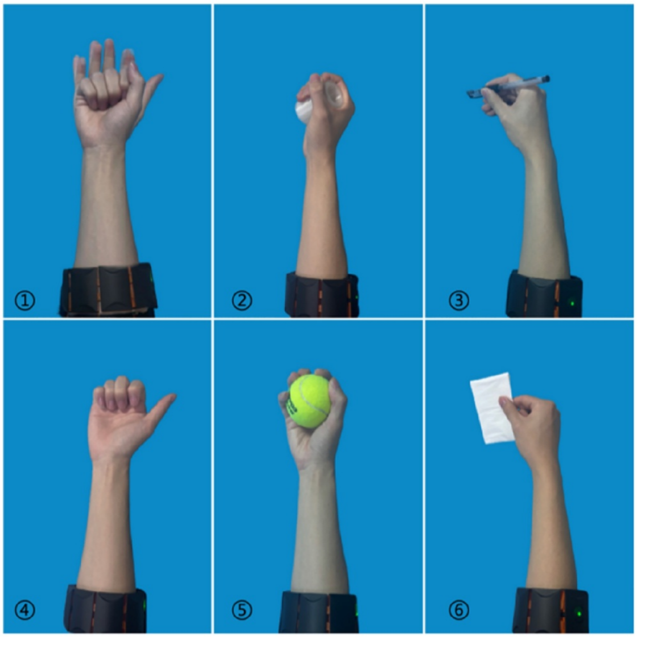

FIGURE 2 | Movement protocols.

2.2.3 | Hand Motion Data Preprocessing

As a non-stationary stochastic signal with primary energy concentrated in 20–150 Hz, EMG exhibits significant inter-subject variability due to factors including perspiration, body hair, and adipose tissue. A Butterworth bandpass filter combined with a 50 Hz notch filter (Figure 3) was applied to mitigate motion artefacts and powerline interference.

This study employs deep learning algorithms to decode finger movement information comprising electromyography (EMG) signals, accelerometer data, and gyroscope measurements. For hand kinematics processing, Kalman Filtering was applied to integrate accelerometer and angular velocity measurements, achieving high-precision orientation angle estimation. The Kalman Filter algorithm Equations (1–5) comprises five sequential stages: (1) state prediction, (2) covariance prediction, (3) Kalman gain computation, (4) state update, and (5) covariance update.

$$\hat{x}_t - = \Phi_t \hat{x}_{t-1} + \omega_t \tag{1}$$

$$P_t^- = \Phi_t P_{t-1} \Phi_t^T + Q_t \tag{2}$$

$$K_t = P_t^- H_t^T \left(H_t P_t^- H_t^T + R_t \right)^{-1} \tag{3}$$

$$\hat{x}_t = \hat{x}_t^- + K_t \left(z_t - H_t \hat{x}_t^- \right) \tag{4}$$

$$P_t = (I - K_t H_t) P_t^- \tag{5}$$

In the Kalman filter process, several key variables and matrices are involved. The predicted state at time t is denoted as $\hat{x}_t^-$, while the state estimate at the previous time step $t-1$ is represented by $\hat{x}_{t-1}$. Φ_t stands for the state transition matrix, and ω_t represents the state noise. The predicted state covariance matrix is labelled as P_t^-, and Q_t is the process noise covariance matrix. K_t refers to the Kalman gain matrix, H_t^T is the transpose of the measurement matrix, and R_t denotes the measurement noise covariance matrix. After the measurement update step, the updated state estimate is expressed as $\hat{x}_t$, and z_t indicates the actual measurement. The updated state covariance matrix is termed as P_t. By fusing the information through the Kalman filter, more accurate attitude angles can be obtained. The motion trajectory of the IMU is shown in Figure 4. This figure illustrates the changes in the roll and pitch angles for the six selected movements. As can be observed from the figure, even though the primary action occurs in the hand, the IMU can still capture subtle changes in the forearm.

Significant scale discrepancies between EMG and IMU sensor outputs were addressed through Z-score standardisation Equation (6), which eliminates dimensional heterogeneity across modalities and enables feature competition on commensurate scales. This preprocessing step approximates data distributions to normality, stabilises gradient directions, accelerates model convergence, and enhances robustness by balancing sensor-specific feature learning. Here, denotes the mean and represents the standard deviation.

$$X_{\text{Standardized}} = \frac{X - \mu}{\sigma} \tag{6}$$

EMG sensors and IMUs have different sampling rates. Over the same time span, EMG signals correspond to 10 time steps, whereas IMU samples correspond to one. To maintain time alignment and prevent information misalignment, the IMU signal is upsampled to 500 Hz from 50 Hz via nearest - neighbour interpolation.

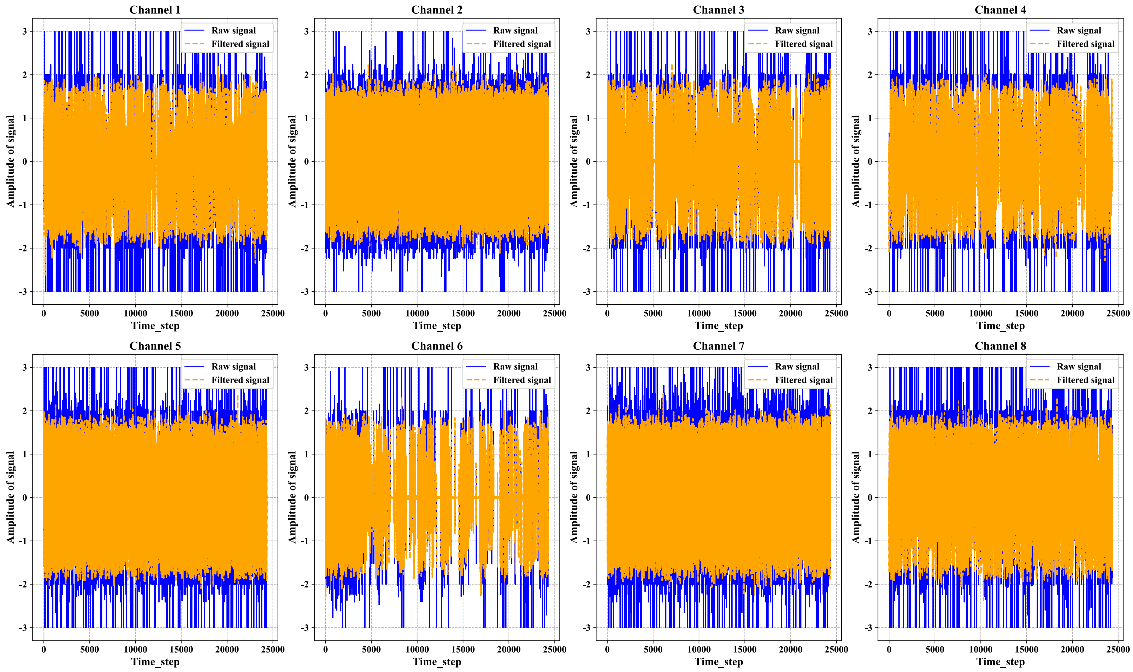

FIGURE 3 | Comparison of each channel before and after filtering.

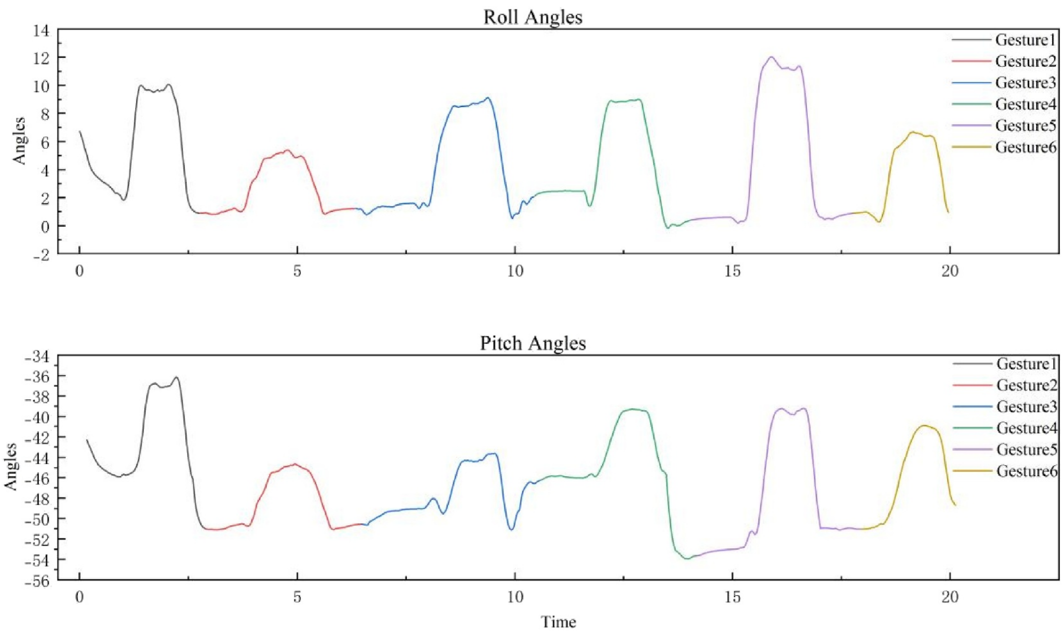

FIGURE 4 | IMU motion trajectory analysis.

2.3 | Model Architecture

A Bi-Stream CNN-Attention-LSTM (BSCAL) neural network processes EMG and IMU sensor data separately to capture their unique spatiotemporal features. EMG signals reflect muscle activity during movement, whereas IMU signals indicate movement posture. By integrating these two types of signals, the model can enhance the robustness of action recognition. The CNN is utilised for extracting local features, the LSTM is designed to capture temporal dependencies from time-series data, and the Attention mechanism helps focus on relevant information by filtering out irrelevant or noisy data. This focus on important information improves the model's generalisation ability by simplifying the model structure and reducing the need for recurrent networks. Moreover, the Attention mechanism can smartly select strongly related features.

2.3.1 | BSCAL Network Configuration

The BSCAL architecture employs dual independent input streams for surface electromyography and gyroscope signals. Each modality-specific branch sequentially processes data through dedicated CNN, attention, and LSTM layers. Distinct convolutional/pooling operations extract localised features, while a custom-designed attention mechanism emphasises critical temporal segments. Temporal dependencies are captured via LSTM structures before fused features are classified through fully connected layers (see Figure 5).

The hybrid dual-stream architecture processes sEMG (input dimension: 10 × 8) and Gyro (10 × 9) signals. The EMG branch incorporates two 1D convolutional stages (Conv1D: 32@5 → 64@3 kernels) with residual connections, max pooling (stride = 2), and batch normalisation for hierarchical spatiotemporal feature extraction. Spatiotemporal attention module amplifies discriminative features before bidirectional LSTM layers (64 → 32 units) capture long-range dependencies. The Gyro branch utilises lightweight single-layer convolution (Conv1D:16@3) and bidirectional LSTM (32 → 16 units). Multimodal features are projected into a unified 64D space via fully connected layers, dynamically fused through learnable gated weights (Sigmoid-activated). The classification head employs two fully connected layers (128 → 64 units) with L1/L2 regularisation ($\lambda_1 = 0.001$–0.0005, $\lambda_2 = 0.002$–0.001) and stepwise Dropout (0.3 → 0.2) to output probabilities for six gesture classes.

2.3.2 | Hybrid Spatiotemporal Attention Mechanism

This study proposes a hybrid spatiotemporal attention mechanism that integrates temporal and spatial attention. The temporal attention computes weights along the time dimension to capture inter-step dependencies, while the spatial attention generates feature-wise weights through convolutional operations to enhance sensitivity to local patterns. A weighted fusion strategy amplifies the model's capacity to prioritise critical temporal segments and discriminative features. The hybrid attention output is formulated as follows:

$$\text{Output} = X \odot \sigma\, \text{Conv1D}(\text{Attention}(\text{Q}, \text{K}, \text{V})) \tag{7}$$

$$\text{Attention}(\text{Q}, \text{K}, \text{V}) = \text{softmax}\left(\frac{\text{QK}^T}{\sqrt{d_k}}\right)\text{V} \tag{8}$$

Here, X denotes the input, σ represents the Sigmoid activation, Conv1D applies a kernel size of 3, $\odot$ indicates element-wise multiplication, and d_k is the feature dimension. The attention mechanism employs global pooling to extract feature statistics, generating channel-wise weights that suppress non-essential features while amplifying critical ones.

2.3.3 | Long Short-Term Memory Networks

The LSTM unit precisely regulates information flow through gating mechanisms (forget gate, input gate, output gate) and cell state interactions, as formalised in Equations (9–14). Learnable gate weights dynamically retain or discard information to capture long-term dependencies. Our architecture combines bidirectional LSTM (BiLSTM) with unidirectional LSTM, where BiLSTM extracts bidirectional contextual dependencies, and the subsequent unidirectional LSTM compresses features while modelling unidirectional temporal dynamics for enhanced performance.

$$f_t = \sigma\left(W_f \cdot [h_{t-1}, x_t] + b_f\right) \tag{9}$$

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \tag{10}$$

$$\tilde{C}_t = tanh(W_C \cdot [h_{t-1}, x_t] + b_C) \tag{11}$$

$$C_t = f_t \otimes C_{t-1} + i_t \otimes \tilde{C}_t \tag{12}$$

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \tag{13}$$

$$h_t = o_t \otimes tanh(C_t) \tag{14}$$

Notations: f_t (forget gate), i_t (input gate), $\tilde{C}_t$ (candidate cell state), C_t (cell state), o_t (output gate), h_t (hidden state). x_t: input vector at step t; h_t: hidden state at t; C_t: cell state at t; W/b: learnable parameters; σ: sigmoid function; $\otimes$: element-wise multiplication.

By processing EMG and IMU data through separate streams, the model captures modality-specific spatiotemporal patterns: EMG reflects muscular activation, while IMU characterises kinematic postures. Sensor fusion enhances gesture recognition robustness. The architecture employs hybrid L1/L2 regularisation and incorporates residual connections in the EMG branch to strengthen feature propagation across deep layers, mitigating gradient vanishing. Detailed EMG branch parameters and output dimensions are listed in Table 1.

2.3.4 | Gated Fusion Mechanism

In the signal fusion component, we employ a gated fusion mechanism to dynamically adjust the contribution of different modalities, as outlined in Equations (15–19). First, E_{proj} and G_{proj} map features from different modalities to a unified dimensional space. Here, $E_{\text{raw}} \in \mathbb{R}^{32}$ represents the raw feature from the EMG branch, while $G_{\text{raw}} \in \mathbb{R}^{16}$ represents the raw feature from the Gyro branch. $W_e \in \mathbb{R}^{64\times32}$ and $W_g \in \mathbb{R}^{64\times16}$ denote the projection matrices. b_e and $b_g \in \mathbb{R}^{64}$ represent the bias terms.

TABLE 1 | EMG branch parameters.

Layer name	Parameters	Output size
Input	/	B × 10 × 8
Conv1	Kernel:5,strides:1	B × 10 × 32
BN,Relu	/	B × 5 × 32
Conv2	Kernel:3, strides:1	B × 5 × 64
BN,Relu	/	B × 2 × 64
Conv3	Kernel:3, strides:1	B × 2 × 64
Shortcut	/	B × 2 × 64
Attention	/	B × 2 × 64
LSTM	Units: 64	B × 32

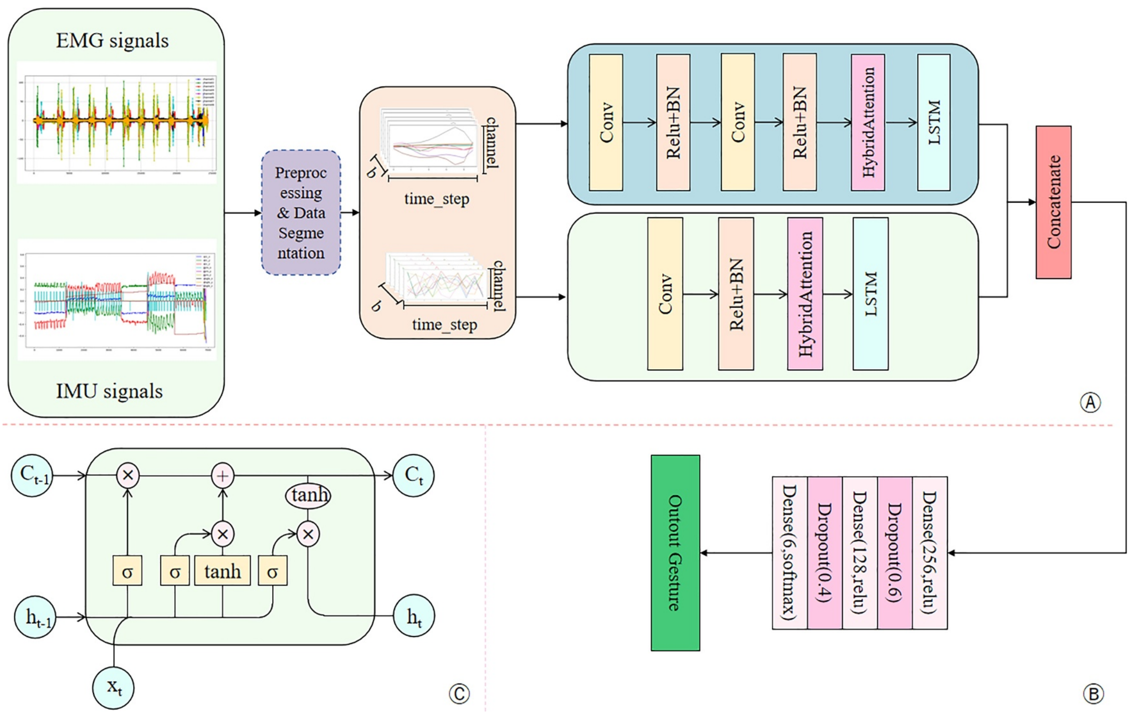

FIGURE 5 | Architecture of the BSCAL network. B: Batch size, c_t: Cell state (memory state), h_t: Hidden state, x_t: Input information.

$$E_{\text{proj}} = \text{ReLU}(W_e \cdot E_{\text{raw}} + b_e) \tag{15}$$

$$G_{\text{proj}} = \text{ReLU}(W_g \cdot G_{\text{raw}} + b_g) \tag{16}$$

Next, we generate gating signals to weight the projected features. $\sigma(\cdot)$ represents the Sigmoid function, which generates feature-level weights. $W^e_{\text{gate}}, W^g_{\text{gate}} \in \mathbb{R}^{64\times 64}$ are the gating weight matrices, and $b^e_{\text{gate}}, b^g_{\text{gate}} \in \mathbb{R}^{64}$ are the gating bias matrices. $\odot$ denotes element-wise multiplication. E_{gated} and G_{gated} represent the results after dynamic weighting. This gating mechanism dynamically adjusts the modality contribution and uses element-wise multiplication for feature selection to enhance important features.

$$E_{\text{gated}} = E_{\text{proj}} \odot \sigma\left(W^e_{\text{gate}} \cdot E_{\text{proj}} + b^e_{\text{gate}}\right) \tag{17}$$

$$G_{\text{gated}} = G_{\text{prog}} \odot \sigma\left(W^g_{\text{gate}} \cdot G_{\text{proj}} + b^g_{\text{gate}}\right) \tag{18}$$

Finally, the weighted features of the two modalities are summed to achieve modality complementarity.

$$\text{merged} = E_{\text{gate}} + G_{\text{gate}} \tag{19}$$

Through the gated fusion mechanism, the model can adaptively learn the feature weights of different sensor signals, thereby improving the classification performance of complex action recognition models.

In this study, the BSCAL model is proposed, which innovatively integrates Convolutional Neural Networks (CNN), spatiotemporal attention mechanisms, and Bidirectional Long Short-Term Memory (BiLSTM) networks to jointly process the spatiotemporal heterogeneity of surface electromyography (sEMG) and Inertial Measurement Unit (IMU) data. Hierarchical CNN structure is used to capture local muscle electrical activity features and kinematic trajectory segments separately. Max-pooling and batch normalisation are employed to enhance feature robustness. Bidirectional LSTM modules (EMG: 64 → 32 units; IMU: 32 → 16 units) are established to parse the long-term temporal dependencies of the signals. A customised spatiotemporal attention module is embedded to strengthen the feature contribution of key channels and times through adaptive weight allocation. A fusion strategy based on gated projection is designed to map bimodal features into a unified embedding space, and dynamic weighted fusion is achieved using Sigmoid-activated gating coefficients.

2.4 | Network Training Strategies

To train and evaluate the model while assessing its generalisation capability, 20% of the data were randomly partitioned as a validation set during the training process. The validation set was randomly selected, ensuring that each epoch utilised an independently chosen validation set, thereby mitigating the risk of overfitting. In this study, the cross-entropy loss function was employed as the loss function as shown in Equation (20).

$$L = -\sum_{i=1}^{N} y_i^{\text{true}} \, log \, y_i^{\text{predict}} \tag{20}$$

where N denotes the number of classes, y_i^{true} represents ground-truth labels, and y_i^{predict} indicates predicted probabilities. The cross-entropy loss function, when combined with a Softmax layer, enables the use of backpropagation for efficient gradient calculation, thereby enhancing both training efficiency and model convergence. To optimise the training process, several strategies are employed: the initial learning rate is reduced; the early stopping patience is extended to 40 epochs; a dynamic learning rate decay strategy is introduced; and the batch size is increased.

Model performance was evaluated using accuracy, precision, recall, and F1-Score metrics:

$$\text{accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{21}$$

$$\text{precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{22}$$

$$\text{recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{23}$$

$$\text{F1-Score} = \frac{2 \times \text{precision} \times \text{recall}}{\text{precision} + \text{recall}} \tag{24}$$

where TP/TN: correctly predicted positive/negative samples; FP/FN: misclassified negative/positive samples.

The F1-Score harmonises precision (ratio of correct positive predictions to total predicted positives) and recall (ratio of correct positives to actual positives), serving as a balanced composite metric. Values approaching one indicate optimal trade-offs between precision and recall.

The confusion matrix [31] provides a tabular evaluation of classification performance, where diagonal elements indicate correctly classified samples (higher values denote better class-wise accuracy), and off-diagonal elements represent misclassifications. This matrix facilitates granular analysis of inter-class discrimination capabilities. Hyperparameter configurations are detailed in Table 2.

3 | Results

In this section, the performance of the proposed BSCAL model is evaluated. First, the experimental results are analysed and compared with those of traditional CNN models, followed by comparisons with other research methods. Subsequently,

TABLE 2 | Hyperparameter configurations.

Hyperparameters	Value
Epoch	200
Batch size	64
Optimiser	Adam
Learning rate	1e-3
Dropout	0.3,0.2
L1+L2 regularisation	0.0005,0.001

ablation studies were conducted to examine the contribution of each module, demonstrating the effectiveness of the BSCAL architecture. Finally, an analysis of performance metrics and attention weights reveals that IMU signals effectively complement muscle activation signals provided by EMG, confirming the efficacy of the multimodal fusion strategy.

3.1 | Experimental Results of the BSCAL Model

This part presents and analyzes the experimental results. The BSCAL model, trained with the parameters delineated in Section 2.3, was tested on the measurement dataset. The loss function and accuracy curves are depicted in Figure 6. To assess the effectiveness of the BSCAL model in gesture recognition, a comparison with a conventional CNN model was carried out using confusion matrices. Figure 7 illustrates the recognition performance of the traditional CNN and the BSCAL models across different gestures.

The BSCAL model has successfully accomplished the task of classifying the gesture dataset. As shown in Figure 6, there is no overfitting in the model for both the training and validation sets, as indicated by the simultaneous decrease and convergence of the loss in both sets. In contrast to the traditional CNN model, Figure 7 reveals that in the CNN model, gesture 2 and gesture 3 are often confused bidirectionally, with 14.7% of gesture 2 being misclassified as gesture 3. This misclassification is primarily due to the highly overlapping motion trajectories and similar EMG activation patterns of the two gestures, making it challenging for IMU and EMG signals to distinguish between them. Both movements activate the flexor digitorum superficialis (FDS), while signal overlap occurs in superficial muscles due to electromyographic crosstalk. Additionally, 15.2% of gesture five is misclassified as gesture 6 due to the proximity of their motion trajectories. Structurally, CNN networks have limited ability to process time series and lack the capacity to capture rapid changes in fine movements. The BSCAL model, however, combines LSTM and Attention mechanisms to handle time series data, thereby addressing the limitations of traditional CNN networks, improving classification performance, and reducing the misclassification rate between gesture 2 and gesture 3 by 13.2%. It also achieves higher accuracy in classifying similar gestures. By integrating CNN with the LSTM and Attention mechanisms, the BSCAL model enhances both classification performance and time - series processing capability.

3.2 | Comparison of the BSCAL Model With Other Research Methods

To comprehensively demonstrate the effectiveness of the proposed BSCAL model, a comparison was conducted with other state-of-the-art methods. Table 3 presents the recognition accuracy of advanced EMG-based classification models. The proposed model achieved an accuracy of 91.83%, with further analysis conducted on accuracy enhancement and statistical significance. Notably, compared to unimodal-based models for gesture recognition, the BSCAL model demonstrated improved recognition accuracy. Although the accuracy improvement over the MV-CNN model is marginal, the BSCAL model exhibits significant structural advantages: whereas MV-CNN requires feature extraction followed by conversion of signals into view-based representations—which increases computational overhead—the BSCAL model directly processes raw signals. Through its deep convolutional and attention modules, it autonomously learns hierarchical feature representations related to gestures and ultimately accomplishes classification tasks, achieving end-to-end feature learning while reducing data processing complexity. Additionally, sensors were integrated during data collection to enhance the wearability for subjects.

3.3 | Ablation Studies

To ensure the generality and representativeness of the ablation study results, five subjects (covering different genders) were randomly selected from the participant pool for model component evaluation. By sequentially removing individual modules and conducting systematic tests based on uniform evaluation metrics, the contribution of each component to the overall performance was analysed. The evaluation results are presented in Table 4.

The first row of Table 4 reflects the performance of the complete model, while the second to fourth rows show the performance when the LSTM, Attention, and CNN mechanisms are removed during training, respectively. By comparing the performance

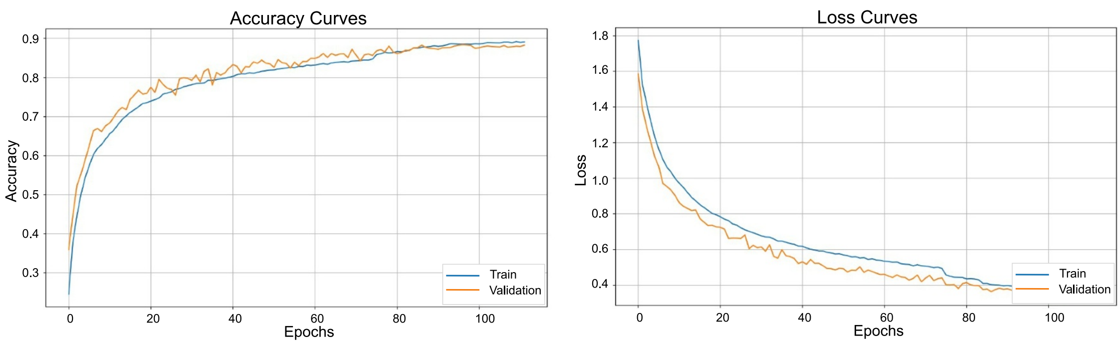

FIGURE 6 | Training curve of the BSCAL model. The left graph shows the accuracy curve of the model and the right graph represents the loss curve of the model.

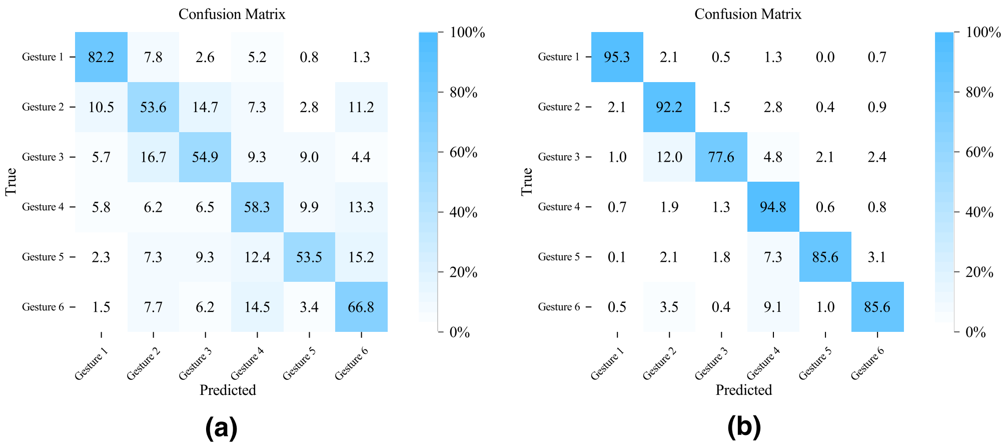

FIGURE 7 | BSCAL network and CNN network confusion matrix. (a) CNN network confusion matrix. (b) BSCAL network confusion matrix.

TABLE 3 | Comparison with other research methods.

Methods	Data type	Accuracy (%)	Improve accuracy (%)	Statistical significance
BSCAL	**EMG + IMU**	**91.83**	—	—
DSC-GRUNet [28]	EMG + ACC	77.48	+14.35	***$p < 0.001$
SE DenseNet [17]	EMG	85.93	+5.9	***$p < 0.001$
SEGTCN [32]	EMG	74.79	+17.04	***$p < 0.001$
Sigimg-GADF-MTF-MSCNN [33]	EMG	88.4	+3.43	**$p < 0.05$
SGRN [34]	EMG	90.22	+1.61	N.s.
MV-CNN [29]	EMG + IMU	90	+1.83	N.s.

Note: All improvements are calculated relative to the BSCAL architecture. The experiment was repeated 10 times. Statistical significance was determined using the *t*-test, where * denotes $p < 0.05$, ** denotes $p < 0.01$, *** denotes $p < 0.001$, and N.s. indicates not significant. The bold values are used to emphasize the performance of our proposed model for easy comparison with other methods.

TABLE 4 | BSCAL model ablation experiment.

Methods	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
BSCAL	**96.76 ± 1.89**	**96.86 ± 1.81**	**97.81 ± 0.76**	**96.76 ± 1.92**
CNN-attention	27.63 ± 1.27	28.43 ± 4.30	29.37 ± 1.19	19.23 ± 0.98
CNN-LSTM	75.88 ± 1.88	76.92 ± 2.99	75.78 ± 1.86	75.36 ± 2.07
LSTM-attention	69.11 ± 2.67	65.17 ± 4.99	67.99 ± 2.35	64.05 ± 2.70

Note: The bold values represent the results of the model's core structure, which serves as the key reference point in our ablation studies.

metrics of the ablated models with the original model, the function of each component can be better understood. The second row shows the model's performance without the LSTM unit. As seen in the table, the accuracy decreases by approximately 70%, highlighting the critical role of LSTM in dynamic time modelling of hand - motion kinesiology, which is particularly important for analysing visually similar gestures. The third row shows the model's performance without the Attention mechanism, resulting in an approximate 21% decrease in accuracy. The combination of temporal and spatial attention enhances the model's feature - selection capability. The last row presents the model's performance without the CNN module, with an accuracy drop of approximately 27%, demonstrating the crucial role of the CNN module in local - feature extraction. Without the CNN module, the model fails to effectively capture local features such as EMG pulse characteristics and motion segments in IMU data, leading to a decline in recognition ability. The ablation studies reveal the hierarchical dependencies among the model's components. Removing any single part disrupts the model's synergy and affects its performance.

3.4 | Comparison of Different Modalities

To demonstrate the impact of multimodal data on classification models, comparative experiments were conducted using both single - modality EMG and multimodal EMG - IMU data (see Figure 8).

By comparing single - modality and multimodal recognition, it is evident that multimodal fusion significantly enhances performance over single - modality approaches. This improvement fundamentally arises from the principle of kinematic - physiological complementarity. While EMG signals are highly sensitive to neuromuscular activation patterns, they do not capture macro - limb movements. Rigid - body motion tracking via IMU data complements this limitation, thereby enhancing the model's robustness. The experimental results demonstrate that multimodal data fusion substantially improves system recognition performance.

3.5 | Visual Analysis of Performance Metrics and Attention Weights

In the context of wearable rehabilitation devices, inference efficiency and memory footprint are critical performance metrics. Validated on both training and testing datasets, the current model achieves a single-inference time of approximately 48 ms and a memory usage of about 0.03 MB across various motion recognition tasks. Although the model integrates CNN, LSTM, and attention mechanisms, resulting in a certain structural complexity, its current performance demonstrates potential for deployment on edge devices. To further enhance deployment suitability, subsequent efforts will focus on optimising the architecture through methods such as model distillation and pruning, thereby better adapting to resource-constrained embedded environments.

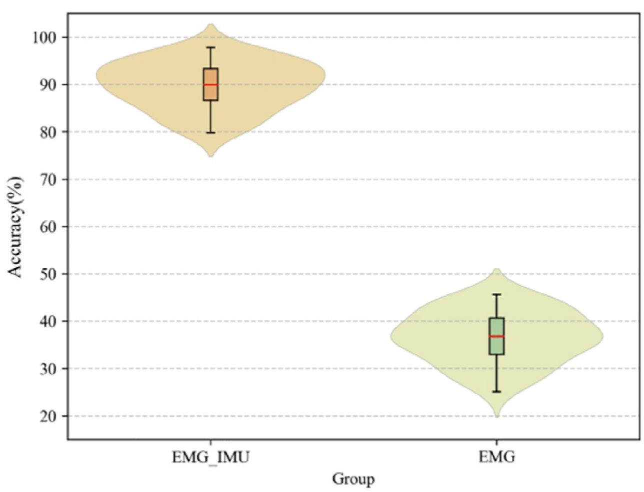

FIGURE 8 | Comparison of single—modality and multimodal EMG and IMU signals with single—modality recognition performance.

Regarding attention weights, a visualisation analysis was conducted. Figure 9 displays the attention weights for EMG signals and IMU signals, respectively.

Figure 9a displays the attention weights of the EMG signals, which range between 0.986 and 1.000. This indicates that the model assigns exceptionally high weights to each motion when processing EMG data, demonstrating its heavy reliance on EMG signals for final classification decisions. From the model's perspective, EMG signals are the most direct and reliable features for distinguishing these six gestures. EMG serves as the decisive modality for motion recognition by the model. Its high and stable attention weights confirm the core role of EMG signals in such fine-grained gesture classification tasks. Figure 9b presents the attention weights of the IMU signals. In contrast to the EMG weights, the IMU attention weights exhibit significant fluctuations, with considerable variations across different motions and attention heads. The importance of different dimensions of IMU data varies greatly depending on the motion. Moreover, the five sets of attention weight patterns shown in Figure 9b are distinct, each focusing on capturing different aspects of the IMU data. For instance, Weight 5 pays more attention to the overall motion amplitude, showing higher weights for Motion 5, while Weight 2 focuses more on specific directional movements or subtle postural changes, displaying greater amplitude variations. IMU serves as an important auxiliary modality for model decision-making, primarily capturing contextual information such as the dynamic process of motions, spatial posture, and force variations. This effectively complements the muscle activation signals provided by EMG. Both EMG and IMU modalities are highly attended by the model, indicating the effectiveness of the multimodal fusion strategy in simultaneously leveraging electromyographic signals and motion dynamics information. The universally high weights of EMG indicate that the model relies more on EMG for fine-grained gesture distinction, while IMU assists in the holistic perception of motion dynamics.

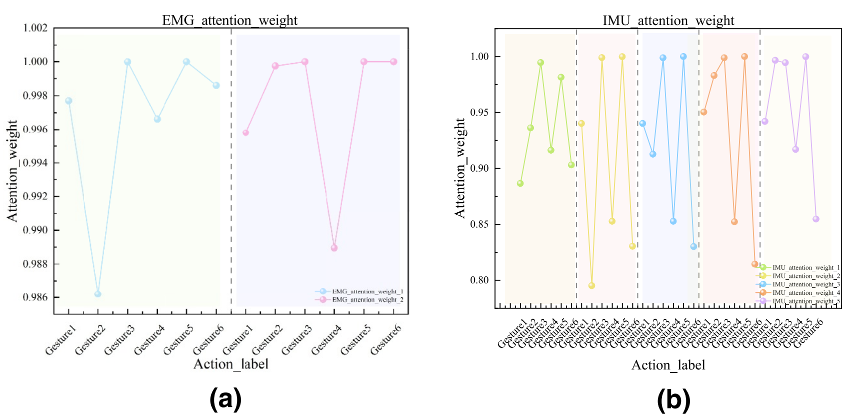

FIGURE 9 | Visualisation of attention signal weights. (a) EMG signal attention weights and (b) IMU signal attention weights.

4 | Discussion

This study proposes a dual - stream neural - network structure based on multimodal heterogeneous data fusion. By integrating EMG and IMU signals, the model significantly improves the accuracy of hand - gesture recognition, achieving an accuracy of 90.41 $\pm$ 1.54%, which is 30% higher than that of conventional CNN networks on average. Compared with existing methods, the proposed model incorporates more types of sensor signals and achieves higher recognition accuracy. It overcomes the limitations of single - modality sensors in classification recognition models and provides a new approach for gesture - recognition systems in wearable devices. Regarding gesture recognition based on surface EMG and IMU signals, ablation studies were used to analyse the roles of the various components and to validate the model's effectiveness. Although the model demonstrates good recognition performance on the dataset, there is still room for improvement. Currently, the dataset used is an offline fixed - gesture dataset, which has certain limitations in recognising complex gestures, and the computational time still needs to be optimised. The attention weights in the IMU module are excessively complex. Model pruning could be applied to remove attention heads with lower weights, thereby reducing model complexity and computational overhead with minimal performance impact. Future work will focus on optimising the BSCAL architecture to achieve a lightweight design while incorporating online learning to enable real-time model feedback. The current model has not yet been validated in patients with stroke or hand motor impairments. This is primarily due to differences between data from healthy subjects and patients (e.g., muscle activation patterns, signal intensity, fatigue levels), which may affect the model's direct performance in clinical settings. The foremost task in future work is to collaborate with clinical institutions to collect patient data and validate and optimise the system in real-world rehabilitation scenarios. In conclusion, this study not only demonstrates the effectiveness of multimodal heterogeneous data fusion in gesture recognition tasks but also lays a theoretical foundation for action recognition methods based on multi-sensor fusion. The proposed dual-stream architecture and fusion mechanism provide a referential framework and methodological support for subsequent researchers to design robust and efficient recognition systems.

Ethics Statement

The study was approved by the Ethics Committee of Northeast Forestry University. All procedures involving human participants were conducted in accordance with the ethical standards of the institutional research committee and with the 1964 Helsinki Declaration and its later amendments.

Consent

Written informed consent was obtained from all the individual participants included in the study.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The datasets generated and analysed during the current study are available from the corresponding author on reasonable request.

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